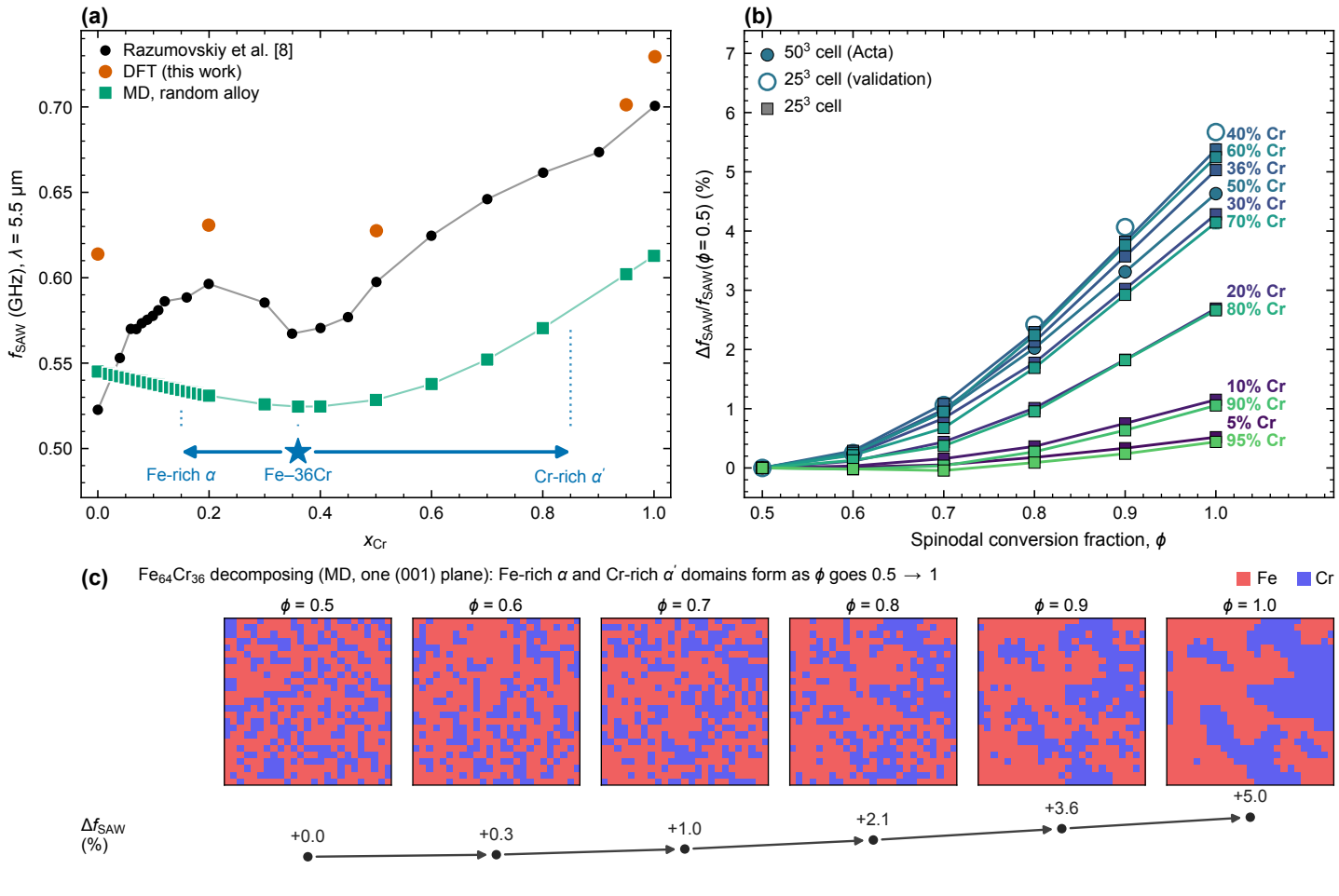

(a)
Razumovskiy et al. [8]
DFT (this work)
MD, random alloy
f_{SAW} (GHz), λ = 5.5 μm
x_{Cr}
Fe-rich α
Fe–36Cr
Cr-rich α′
(b)
50^3 cell (Acta)
25^3 cell (validation)
25^3 cell
$\Delta f_{SAW}/f_{SAW}(\phi = 0.5)$ (%)
Spinodal conversion fraction, ϕ
40% Cr
60% Cr
36% Cr
50% Cr
30% Cr
70% Cr
20% Cr
80% Cr
10% Cr
90% Cr
5% Cr
95% Cr
(c)
$Fe_{64}Cr_{36}$ decomposing (MD, one (001) plane): Fe-rich α and Cr-rich α′ domains form as ϕ goes 0.5 → 1
Fe
Cr
ϕ = 0.5
ϕ = 0.6
ϕ = 0.7
ϕ = 0.8
ϕ = 0.9
ϕ = 1.0
Δf_{SAW} (%)
+0.0
+0.3
+1.0
+2.1
+3.6
+5.0